

The Response Gap

How Algorithmic Credit Engineers Forced Urgency and Misattributes
Loss — and How Mathematics Can Correct It

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Epistemic-status note — a discipline this paper both proposes (§8.3) and practices. Claims are marked [documented] (public record), [modeled] (within a stated model), [prospective] (testable forward claim), or [open] (unresolved, flagged not asserted). The two words in the title — “forced” and “misattributes” — are held to this standard explicitly in §6.

Abstract

A class of algorithmic credit and margin mechanisms — automated collateral calls, forced-liquidation triggers, risk-based repricing — can, under a stated model, produce loss through engineered time pressure (the *response gap*), while the deploying institutions frame the loss as the borrower’s poor judgment (the *attribution gap*) rather than a designed outcome. The response gap is the interval Δt between automated breach-detection and irreversible action — a window that can be shorter than an informed response requires, set by the mathematics of the trigger rather than by circumstance. This paper defines the gap as an auditable governance object, situates it against the established literatures whose parts it assembles (fire sales, DeFi liquidation extraction, algorithmic-lending fairness), examines honestly *where the “forced” and “misattributed” claims are evidenced and where they are not* (§6), shows how the gap compounds with asymmetric rate structures into a repeatable wealth-transfer channel [modeled], and predicts that loss-minimizing model fine-tuning will *tighten* the gap absent an explicit response-time constraint [prospective]. It then argues that the same formal tools can close the gap, proposing four correctives grounded in existing precedent, and closes with policy recommendations. The central move is that the harm’s *unmeasurability under current opacity is itself the governance failure*; the correctives double as the instruments that would let it be measured.

Keywords: response gap · forced urgency · algorithmic credit · forced liquidation · margin call · wealth transfer · auditable AI · formal verification · GELSI · fair lending

1 · Introduction — two failures dressed as one narrative

Institutions extending leveraged credit — margin lending, collateralized loans, algorithmic trading platforms, and increasingly AI-underwritten consumer credit

— share a structural feature: a model continuously monitors the ratio of a borrower’s collateral to their obligation, and when that ratio breaches a threshold, the system acts, often without meaningful human review and within a timeframe set by the model rather than by the borrower’s capacity to respond.

The industry narrative around such events is consistent: the borrower took excessive risk, failed to maintain margin, ignored warnings. This paper argues that two distinct failures may hide inside that single story. The first is a **mechanism failure**: the mathematics of the trigger is designed — deliberately, or through unexamined design inheritance — to minimize the borrower’s window for informed response. The second is a **narrative failure**: the institution that built and benefits from the mechanism frames the resulting loss as the borrower’s moral failing. Where both hold, they reinforce each other — the narrative failure makes the mechanism failure invisible by relocating the causal story from the system to the individual.

A word on epistemic honesty up front, because it is the paper’s method as well as its subject. That the borrower-blaming *narrative* occurs is documentable [documented]. That the loss is genuinely *forced* (constraint-driven rather than chosen), and therefore that the blame is a *misattribution*, is — in the general case — *not* directly observable, for a reason the paper itself proves (§3, non-identification). This paper therefore does not assert a proven harm. It defines an auditable object, shows the domains where the forcing *is* observable, and specifies what evidence would settle the rest (§6).

2 · Related work — the parts are known; the bridge is the contribution

The mechanism described here is not an undiscovered phenomenon. Its components are studied across three literatures that seldom cite one another; the contribution is the bridge between them and the governance object it isolates. Stating this plainly is the honest and stronger position.

Macro-finance. Forced-liquidation loss and transfer are foundational: Shleifer & Vishny on asset fire sales (1992; and *Fire Sales in Finance and Macroeconomics*, 2011); Brunnermeier & Pedersen on the margin and loss spirals of *Market Liquidity and Funding Liquidity* (2009), whose feedback this paper’s cascade restates; and their *Predatory Trading* (2005), which already models agents deliberately forcing others to liquidate at fire-sale prices [documented]. What the response gap adds: the fire-sale literature is about *price feedback*; it holds fixed the variable isolated here — the borrower’s *time to respond* between detection and irreversibility.

Decentralized finance. A recent empirical literature studies liquidation extraction directly: Qin et al., *An Empirical Study of DeFi Liquidations* (ACM IMC, 2021); *Toxic Liquidation Spirals* (2022), which — relevant to §8.4 — already proposes halting liquidations past a loan-to-value threshold; and *reversible call option* designs (Miqado) that defer liquidation [documented]. What the response gap adds: §8’s correctives *generalize* these DeFi-native proposals to regulated consumer and AI-mediated credit, and treat the response window as a *disclosed regulatory parameter* rather than a protocol design choice.

Algorithmic-credit governance. Opacity, attribution, and disparate impact of AI underwriting are active in law and FAccT: model-reporting standards (Mitchell et al., *Model Cards*, 2019); less-discriminatory-algorithm work (Gillis, Meursault

& Ustun, FAccT 2024); CFPB guidance requiring specific, accurate adverse-action reasons for AI credit decisions; and enforcement (the 2024 Apple Card action against Apple and Goldman Sachs; the 2025 Massachusetts settlement over an AI underwriting model) [documented]. What the response gap adds: this literature targets *who* is denied/mispriced and *why*; it does not target *when* — the timing of an automated liquidation and the borrower’s window to act.

AI safety. The §7 prediction is an instance of reward/objective misspecification long noted in AI safety (e.g., Amodei et al., *Concrete Problems in AI Safety*, 2016). The contribution is the specific, testable form the failure takes in credit *timing*.

Net of prior art, new here are: (i) the cross-silo synthesis; (ii) the response gap as a first-class, disclosed, bounded regulatory parameter; (iii) the testable fine-tuning prediction; and (iv) the epistemic-disclosure corrective from formal verification.

3 · Forced urgency and its two variants

Let a borrower or holder face a non-deferrable obligation $\bar{c}$ (rent, mortgage, a margin requirement) whose breach carries a default cost χ . Define **urgency** as λ , the shadow price (Lagrange multiplier) on the liquidity constraint, strictly positive exactly when $\text{cash} + \text{income} + \text{credit} < \bar{c}$. A party at $\lambda > 0$ rationally accepts *any* positive price or term, so a loss realized at $\lambda > 0$ carries no information about that party’s preferences [modeled]. Call this condition **forced urgency**. Two variants exploit it:

- **The attribution gap** (*epistemic* variant; companion paper). Because λ is latent to any outside observer, a loss *caused* by constraint is *attributed* to the borrower’s preferences or judgment. Its signature is a non-identification result: on trades, prices, and paper gains with λ unobserved, “the borrower chose badly” and “the borrower was constrained” are **observationally equivalent**; identification requires an instrument that shifts liquidity independently of returns.
- **The response gap** (*temporal* variant; this paper). An automated system compresses the interval Δt between breach-detection and irreversible action below the time an informed party would need to respond. Its signature is engineered timing.

The connecting claim: **the response gap manufactures the very $\lambda > 0$ moment that the attribution gap then misreads**. Note the honest consequence, carried into §6: because identification requires an instrument, neither “forced” nor “misattributed” is directly readable from ordinary data — except where the trigger is mechanical and the timing observable.

4 · Formalizing the response gap

Define the **response gap** as the interval Δt between (a) detection of a threshold breach and (b) the moment the resulting forced action becomes irreversible, measured against the minimum time a reasonably diversified, reasonably informed borrower would require to cure the shortfall or make a considered decision.

The gap is frequently a *designed parameter*, not a physical necessity [modeled]. Collateral thresholds set close to prevailing volatility bands, liquidation windows measured in minutes on some algorithmic platforms, and margin-call notices routed through channels a borrower may not monitor in real time are choices.

Risk modeling (Value-at-Risk thresholds, collateralization ratios, stress triggers) can carry a response-time budget as an explicit variable; that it typically does not is a design decision with distributive consequences, not a neutral constraint.

The gap is **asymmetric by construction**: the institution setting the threshold has full advance knowledge of where it sits and how liquidation executes; the borrower typically does not know precisely when a breach will trigger action until it has. This information-and-timing asymmetry has a direct mathematical description, and it — where present — produces realized loss the borrower could not have avoided [modeled].

5 · The wealth-transfer channel — asymmetric rate structures

The response gap compounds with a second asymmetry: the spread between what a lender charges a borrower under duress and what it pays for the capital it holds or redeploys. When a forced sale occurs, the liquidated collateral is typically absorbed or resold at prevailing — often distressed — prices, depressed in part by the forced sale itself. The party that set the narrow window is *often, though not always*, positioned to benefit: by repricing residual exposure punitively, by acquiring the asset below pre-crisis value, or by collecting a liquidation fee or bonus [modeled]. This is a tendency, not a universal; counter-cases exist where the liquidator did not capture the value (Archegos 2021, losses on the banks; Celsius 2022, the platform itself insolvent) [documented].

In a limited-arbitrage setting the price satisfies $p = \mu - \kappa S$, where forced supply S rises as price falls; the loop amplifies loss by $A = 1/(1 - \rho)$, $\rho = \kappa |\partial S / \partial p|$, diverging into fire-sale equilibria beyond a threshold [modeled]. Modeled thus, each forced-liquidation event is, for a well-capitalized institution, a favorable-odds transaction priced in advance; for the borrower it is a single, high-variance, poorly understood tail event. The rate asymmetry is then not merely a byproduct of risk pricing but a channel converting a *timing* advantage into a revenue stream that scales with the number of borrowers pushed through response gaps rather than with aggregate risk borne — a **governance** question, since it operates through information architecture and timing design borrowers cannot audit or contract around.

6 · Evidence, identification, and what is not yet shown

This is the section the paper’s own logic demands, and the one it must not skip. Section 3 proves that, from ordinary data, “forced” and “chosen” are observationally equivalent. The disciplined position is therefore: mark the general claims [modeled], anchor evidence where the trigger makes forcing observable, and specify the tests that would settle the rest.

6.1 Where “forced” is directly observable — mechanical triggers. In margin engines and, most cleanly, in **DeFi lending**, forcing is *definitional*: a contract executes liquidation at a threshold breach, not by the borrower’s choice, and in DeFi the trigger, the timing, and the realized loss are all **on-chain and measurable**. Qin et al. (2021) document liquidation incentives and instabilities empirically; MEV research documents value extracted at borrowers’ expense. DeFi is thus the natural laboratory in which the entire mechanism — response gap, forcing, wealth transfer — is measurable *today*, without new disclosure [documented].

6.2 Identification by instrument — where the trigger is not mechanical.

In general asset markets, forcing is inferred through instruments that shift liquidity independently of returns. Coval & Stafford (2007) identify forced sales from mutual-fund redemption flows unrelated to fundamentals — precisely the instrument strategy §3 requires. The companion attribution-gap paper marshals liquidity-shock natural experiments (2020, 2022) that reject a preference-only account. This paper *cites* that identification rather than re-asserting forcing [documented, indirect].

6.3 The attribution claim. That a borrower-blaming narrative follows these events is documentable from public and regulatory commentary [documented]. That the narrative is a *misattribution* is an inference conditional on 6.1–6.2, and is [modeled] outside the mechanical cases. The paper claims the first plainly and the second conditionally.

6.4 What is not yet shown — the measurement agenda. The honest gaps, stated as tests: (i) *In DeFi, now*: does a shorter response window / larger liquidation penalty predict greater borrower loss, controlling for volatility? On-chain data can answer this directly. (ii) *Under disclosed Δt (post-corrective)*: does realized loss rise as Δt falls, holding risk fixed? (iii) *The §7 prediction*: on a fixed model class and dataset, does a training objective lacking a response-time term select tighter Δt than one containing it? Testable on held-out or synthetic policies. (iv) *The meta-point*: that (ii) cannot be run today, because Δt and λ are undisclosed, **is** the governance failure. The absence of general evidence is not evidence of absence; it is the designed opacity the correctives (§8) exist to remove. The correctives are therefore also the instruments.

7 · Where AI fine-tuning enters

As institutions deploy machine-learning underwriting fine-tuned on historical default and liquidation data, the response gap risks being *further optimized rather than closed*. A model trained to minimize institutional loss, with no explicit constraint representing the borrower’s fair opportunity to respond, will tend to learn timing parameters that *tighten* Δt , because a narrower gap reduces exposure to adverse price movement between detection and execution. Absent deliberate governance, fine-tuning on institutional-loss-minimizing objectives will systematically and silently worsen the response gap — not through malice but through ordinary optimization pressure on an incompletely specified objective (reward misspecification applied to credit timing). This is a **direct, testable prediction** [prospective]: fixing a lender’s model class and data, an objective lacking a response-time term will select tighter Δt than one containing it, measurable on held-out policies. Its implication: any framework for AI-mediated credit must require *response-time-to-borrower* as an explicit, disclosed, bounded parameter in the model specification, not an emergent property left to the optimizer.

8 · Correctives — closing the gap mathematics opened

The same formal tools responsible for the response gap can, under different constraints, close it. Four correctives, each grounded in existing precedent.

8.1 Transparent, published liquidation formulas. Thresholds disclosed as explicit, auditable formulas at contract time, not merely referenced by policy language — mirroring disclosure already common in some derivatives markets.

8.2 Mandatory minimum response-time budgets. By analogy to the rule-based, non-discretionary finality of distributed-ledger protocols, and directly extending DeFi *reversible-liquidation* proposals, credit systems should carry a minimum, disclosed Δt between detection and irreversible action, calibrated to a defined “reasonable response” standard rather than the institution’s risk-minimizing preference alone. Import the design *principle*, not the technology.

8.3 Epistemic-disclosure standards for underwriting models. Borrowing from formal-verification practice — where each claim is tagged by epistemic status (proved, cited, conjectural, open) — AI-underwriting systems should disclose the status of the parameters governing a borrower’s liquidation risk: which thresholds are empirically validated on the institution’s own data, which are inherited defaults, and which are optimizer-selected without human-legible justification. This extends the model-card and adverse-action lineage to the *timing and threshold* parameters. (This paper adopts the same tagging for its own claims; the standard should bind the critic first.)

8.4 Rule-based circuit breakers decoupled from institutional discretion. Exchange-level circuit breakers, and DeFi threshold-halting proposals, show that automatic, non-discretionary pauses can prevent cascading forced-sale spirals. Extended to the account level: an automatic pause triggered by the size or speed of a proposed forced sale, adjudicated by a rule fixed and disclosed in advance, not by the liquidating institution’s real-time judgment.

9 · Policy implications

The contribution to the GELSI literature is threefold: a **formal definition of the response gap** as an auditable governance object rather than a diffuse complaint about “predatory lending”; the specific, **testable prediction** that loss-minimizing fine-tuning worsens the gap absent intervention [prospective]; and correctives grounded in **existing technical precedent** rather than novel regulatory infrastructure. Concretely, regulators — in the U.S., a coordinated standard across the CFPB, OCC, and state banking regulators; internationally, coordination through bodies such as the FSB — should adopt minimum response-time-budget and epistemic-disclosure requirements as a condition of deploying automated or AI-mediated liquidation systems, and should mandate the disclosure that would make the §6 tests runnable. Mathematical transparency is positioned not as a constraint on financial innovation but as its necessary corrective — and as the instrument that turns an alleged harm into a measured one.

10 · Conclusion

Forced urgency — a borrower at a positive liquidity shadow price who will accept any terms — can be exploited two ways: an attribution gap that misreads the resulting loss as a preference, and a response gap that engineers the loss by compressing the borrower’s time to act. The paper does not claim this harm is proven in general; its own non-identification result forbids that from ordinary data. What it claims is narrower and firmer: the mechanism is real and *measurable where triggers are mechanical* (DeFi, margin), it is *identifiable by instrument* elsewhere, the borrower-blaming narrative is *documented*, and the reason the general case cannot yet be measured is the designed opacity the correctives remove. The honest framing is the strongest one: not a proven indictment, but an auditable object,

a measurement agenda, and a set of correctives that are also the instruments — built openly on prior parts.

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Companion works

- **The attribution gap** — *The Forced Urgency Gap* (WP-32): forced urgency and the λ non-identification result, with the 2020 and 2022 liquidity-shock natural experiments. The epistemic variant of the mechanism developed here.
- **“Algorithmic Urgency”** — Book 6, *Principia Orthogona*, WP-56.
<https://totogt.github.io/geometry/book6/wp56-algorithmic-urgency.html>
- **Positional Dominance under Non-Contestability** — [10.5281/zenodo.21753025](https://zenodo.org/record/21753025). Two-player stochastic network game; hub control dominates velocity investment above $\sigma^* \approx 1/3$; constants machine-checked in Lean 4.

Original working paper, version 1.

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